Mapping short reads

Tools and quality control

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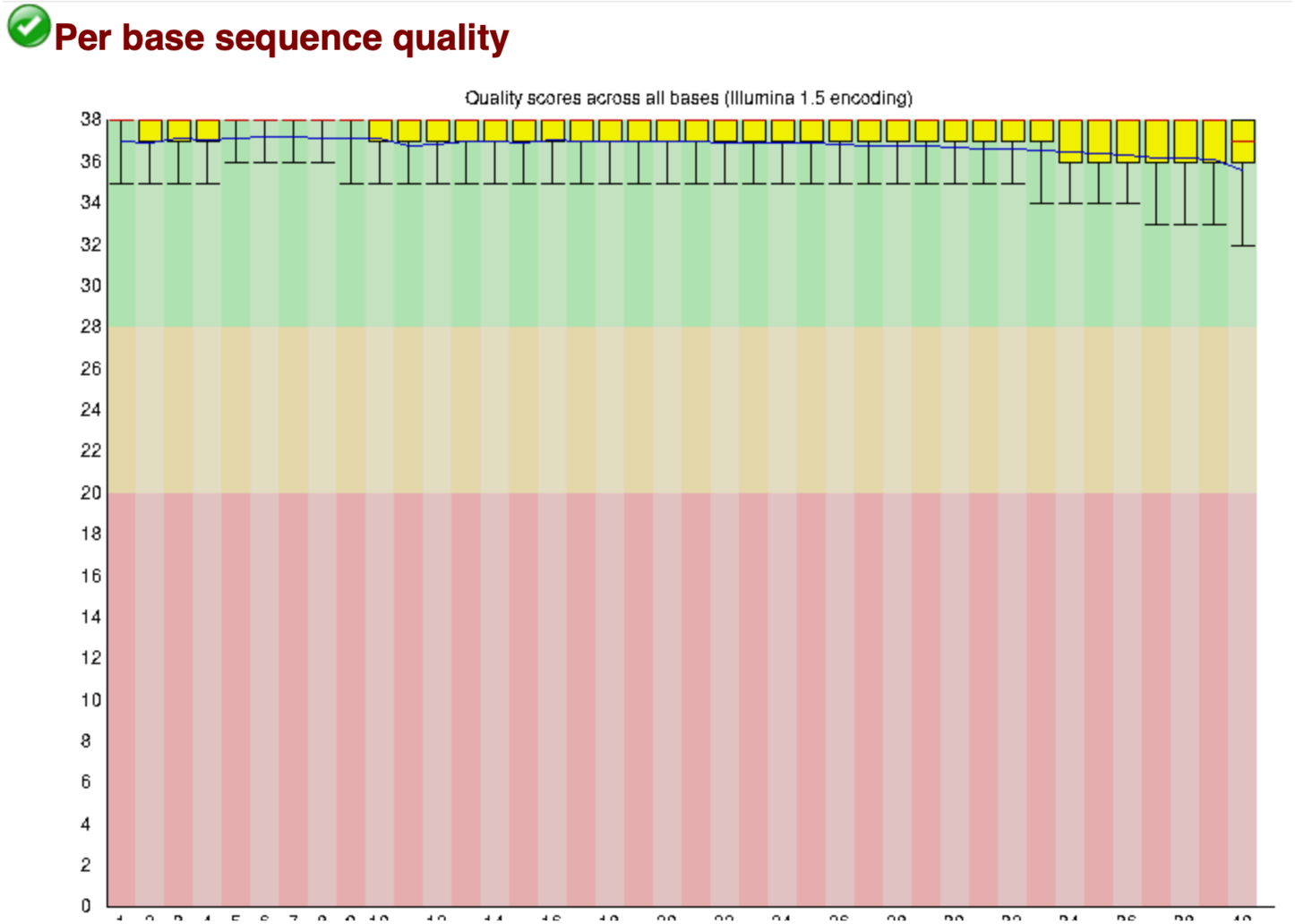
Learning outcomes

- **Quality control of reads**
  - Common issues and how to solve them
- **Understand reference-based alignment basics**
  - Software to use
  - Quality control
- **Advantages and disadvantages compared to *de novo* assembly**

Quality control

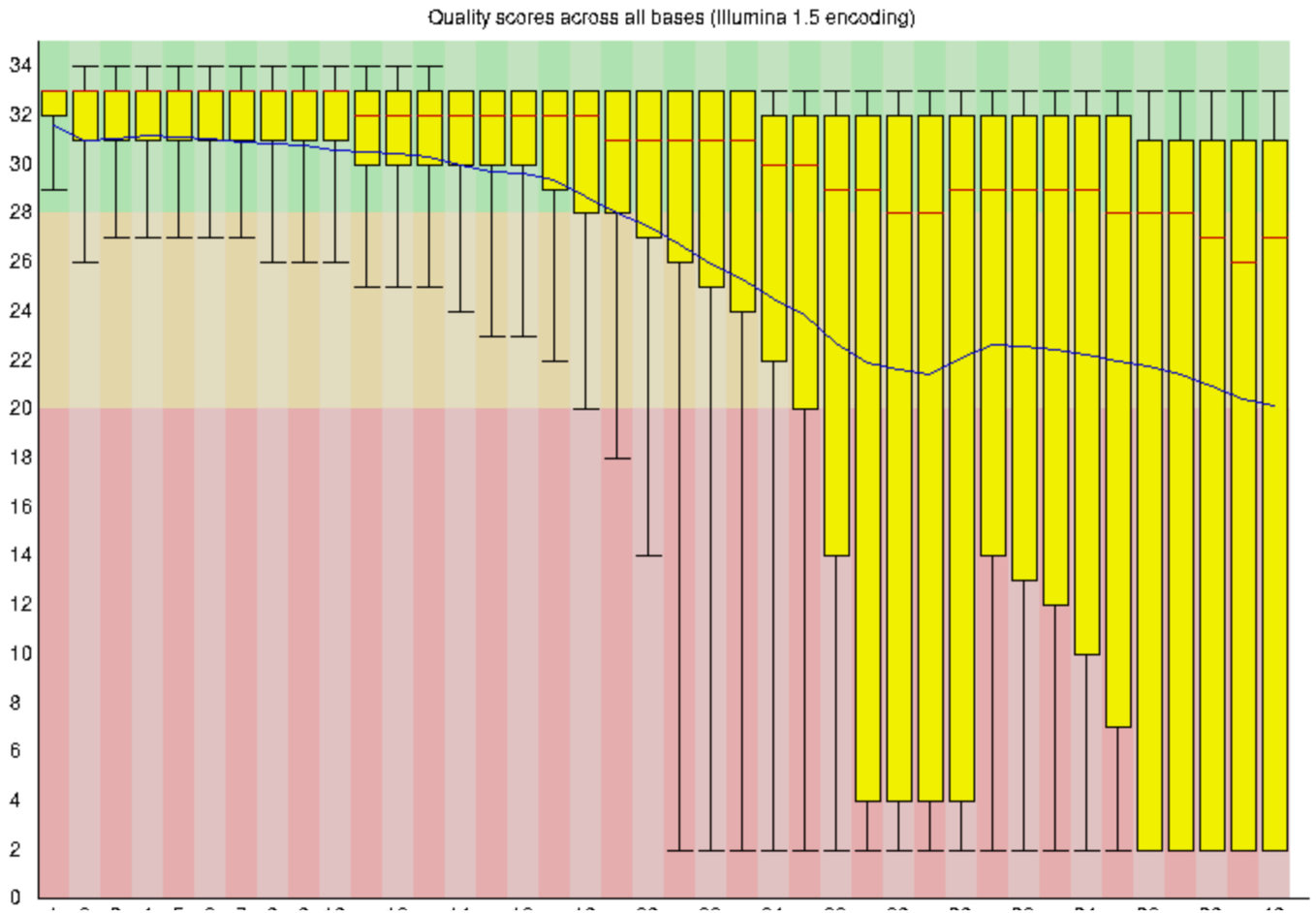
FASTQ reads with FastQC (Babraham Institute)



Quality control

FASTQ reads with FastQC (Babraham Institute)

Per base sequence quality



Quality control

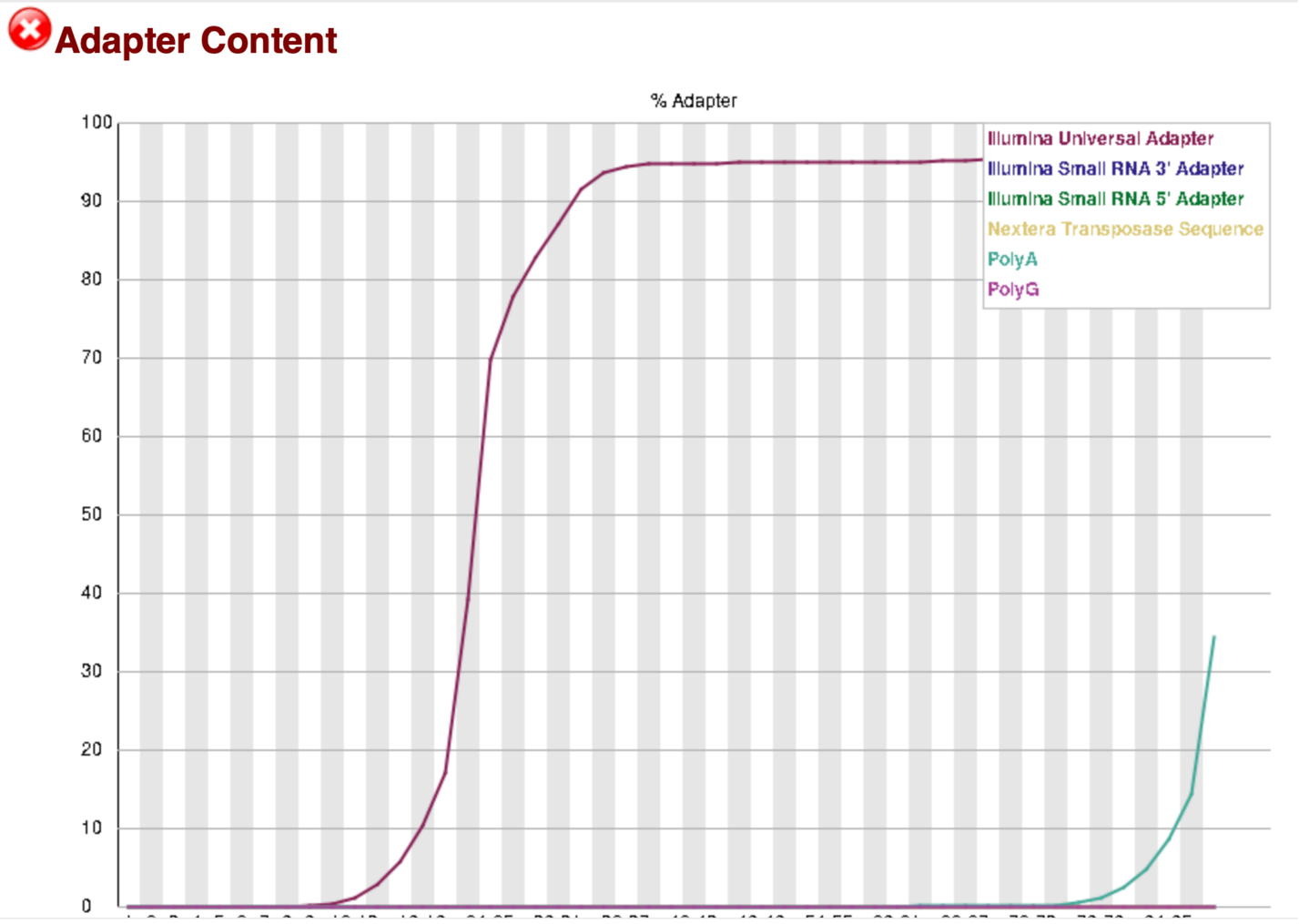
FASTQ reads with FastQC (Babraham Institute) - adapter dimer contamination

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCT	8122	8.122	Illumina Paired End PCR Primer 2 (100% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGATCGGAAG	5086	5.086	Illumina Paired End PCR Primer 2 (97% over 36bp)
AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTAC	1085	1.085	Illumina Single End PCR Primer 1 (100% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGAAG	508	0.508	Illumina Paired End Sequencing Primer 2 (100% over 36bp)
AATTATACGGCGACCACCGAGATCTACACTCTTTCCCTAC	242	0.242	Illumina Single End PCR Primer 1 (97% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAAGATCGGAA	235	0.23500000000000001	Illumina Paired End Adapter 2 (96% over 31bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCGAGATCGGAAGA	228	0.22799999999999998	Illumina Paired End Adapter 2 (96% over 28bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACG	205	0.20500000000000002	Illumina Paired End Sequencing Primer 2 (100% over 36bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGTCGGAAG	183	0.183	Illumina Paired End Adapter 2 (100% over 32bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGATCGGAA	183	0.183	Illumina Paired End Adapter 2 (100% over 32bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGAACT	164	0.164	Illumina Paired End PCR Primer 2 (97% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGTCT	129	0.129	Illumina Paired End PCR Primer 2 (97% over 40bp)
AATTATACTTCTACCACCTATATCTACACTCTTTCCCTAC	123	0.123	No Hit
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACT	122	0.122	Illumina Paired End Sequencing Primer 2 (100% over 36bp)
CGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGC	113	0.11299999999999999	Illumina Paired End PCR Primer 2 (96% over 25bp)

Quality control

FASTQ reads with FastQC (Babraham Institute) - small RNA with read-through adaptor



Quality control

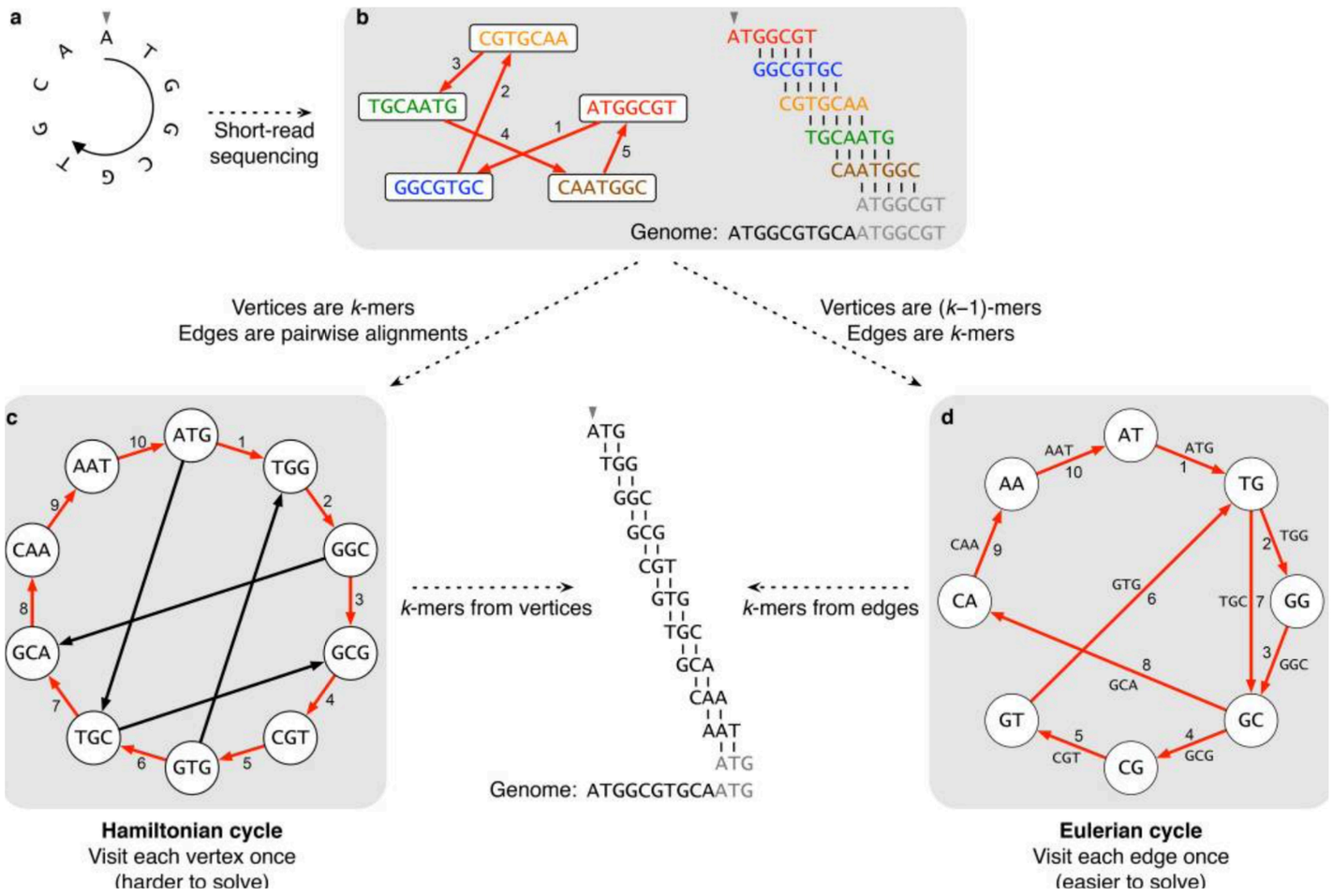
Adaptor trimming reads

- Tools such as ‘cutadapt’ and ‘trimmomatic’ can be used to remove adaptor and/or overrepresented sequences
  - Trimmomatic will also cut low quality sequences

De novo assembly

No reference required

- Converts reads into a detailed set of sequences corresponding to the chromosome of an organism
- Needed when sequencing an organism for the first time
- Longer reads better to resolve repetitive regions
  - SPAdes is a short read assembler
    - Capable of hybrid assembly
  - Canu uses ONT long reads
  - Hifiasm uses PacBio HiFi long reads



Compeau, P.E., *et al. Nat Biotechnol* **29(11)**, 987-991 (2011)

Assembly vs. alignment

Pros and cons

- Pros:
  - Assembly - find structural variants and large copy number variation (CNV)
  - Alignment - good for small variants (indels, SNPs, CNVs)
- Cons:
  - Assembly - loose information about variants or depth of coverage; difficult to resolve repetitive regions without long reads
  - Alignment - errors/false positive variants if high levels of repetitive regions

Mapping reads to reference genome

Like a jigsaw puzzle

- Alignment is generally quicker and computationally less expensive than assembly, as we have a guide (the reference)
- Burrows-Wheeler algorithm (BWA) reconstructs the short read data in our FASTQ files using the reference genome as a guide
  - BWA ‘mem’ algorithm is the fastest and most accurate in the BWA package, and can handle short reads up to 250 bp.

```
bwa mem –M reference.fa R1.fastq.gz R2.fastq.gz > isolate_name.sam
```

The SAM file format

Sequence Alignment/Map

- Binary version = BAM
  - convert using samtools: `samtools view -bS isolate_name.sam > isolate_name.bam`
- Basic alignment!
  - PCR duplication not considered, no co-ordinate sorting, no indel consideration

(1) The query name of the read is given (M01121...)

(2) The flag value is 163 (1+2+32+128)

(3) The reference sequence name, chrM, refers to the mitochondrial genome

(4) Position 480 is the left-most coordinate position of this read

(5) The Phred-scaled mapping quality is 60 (an error rate of 1 in 10^6)

(6) The CIGAR string (148M2S) shows 148 matches and 2 soft-clipped (unaligned) bases

```
home/bioinformatics$ samtools view 030c_S7.bam | less
M01121:5:000000000-A2DTN:1:2111:20172:15571     163     chrM
480     60      148M2S  =       524     195     AATCTCATCAAT
ACAACCCTCGCCCATCCTACCCAGCACACACACACCGCTGCTAACCCCATACCCCGAACC
AACCAAACCCCAAAGACACCCCCCACAGTTTATGTAGCTTACCTCCTCAAAGCAATAACC
TGAAAATGTTTAGACGGG  BBBBBFFB5@FFGGGFGEGGGEGAAACGHFHFEGGAGFFH
AEFDGG?E?EGGGFGHFGHF?FFCHFH00E@EGFGGEEE1FFEEEHBGEFFFGGGG@</0
1BG212222>F21@F11FGFG1@1?GC<G11?1?FGDGGF=GHFFFHC.-
RG:Z:Sample7    XC:i:148        XT:A:U  NM:i:3  SM:i:37
AM:i:37 X0:i:1  X1:i:0  XM:i:3  XO:i:0  XG:i:0  MD:Z:19C109C0A17
```

(7) An = sign shows that the mate reference matches the reference name

(8) The 1-based left position is 524

(9) The insert size is 195 bases

(10) The sequence begins AATCT and ends ACGGG (its length is 150 bases)

(11) Each base is assigned a quality score (from BBBBB ending FHC.-)

(12) This read has additional, optional fields that accompany the MiSeq analysis

Flag Value	Meaning	Flag Sum
1	read is paired	1
32	read2 was reverse complemented	33
64	read1	97
2048	Supplementary alignment	2145

Improving the alignment

Co-ordinate sort

- `samtools sort isolate_name.bam isolate_name.sorted`
- All BAM files need to be indexed, which creates a new file allowing faster look-up of data
  - `samtools index isolate_name.sorted.bam`
  - This creates a file with suffix ‘bam.bai’

Improving the alignment

Fix the read groups

- We use ‘AddOrReplaceReadGroups’ from Picard to ‘fix’ the BAM file.
  - Downstream analyses (e.g. GATK) require the BAM file to contain read group information:
    - Read group identifier - this is the sequencer flow cell, lane number and name.
    - Read group platform - e.g. Illumina
    - Read group library - this is needed for marking duplicates to determine which read groups contain molecular duplicates
    - Read group platform unit - this is the run ‘barcode’ or the adaptor sequence
- The command may look a bit like this:

```
picard AddOrReplaceReadGroups -I isolate_name.sorted.bam -O isolate_name.fixed.sorted.bam --SORT_ORDER coordinate
--RGID K00166 --RGLB dnaseq --RGPL illumina --RGSM ’WGS’ --CREATE_INDEX TRUE --RGPU unknown --VALIDATION_STRINGENCY
SILENT
```

Improving the alignment

Marking duplicates

- Use Picard to mark any duplicated reads due to sequencing errors to prevent them being included in variant calling

```
picard MarkDuplicates -I isolate_name.fixed.sorted.bam -O
isolate_name.sorted.marked.bam --CREATE_INDEX TRUE --METRICS_FILE
picard_info.txt --REMOVE_DUPLICATES false --ASSUME_SORTED true --
VALIDATION_STRINGENCY SILENT
```

- This is still a rough ‘global’ alignment
- The rest of the pipeline is entirely GATK for improving the alignment further and variant calling

Improving the alignment

Base Quality Score Recalibration

- BQSR is a pre-processing step that detects systematic errors made by the sequencing machine when it estimates the accuracy of each base call.
  - BQSR adjusts locally according to the quality of each base call
  - Variant callers rely heavily on the quality score assigned to individual bases
- Two passes required
  - First pass generates a recalibration table using a ‘known sites’ vcf of SNPs and indels (these are not gold standard!) using the ‘BaseRecalibrator’ tool
  - Second pass uses ‘ApplyBQSR’ to adjust the base quality scores based on the model, producing the final BAM file
- Recommended to build two models i.e. run this twice for quality control

Improving the alignment

Base Quality Score Recalibration - commands

- First we need to provide a rough guide of variants (SNPs and indels) for BQSR. HaplotypeCaller (GATK) calls variants, and by default assumes a diploid organism so we need to specify ploidy is 1 (haploid)

```
gatk HaplotypeCaller –R reference.fa –I isolate_name.sorted.marked.bam –ploidy 1 –O
isolate_name.raw_variants.vcf
```

- This file contains both SNPs and indels, and can be used for BQSR:

```
gatk BaseRecalibrator –R reference.fa –I isolate_name.sorted.marked.bam –-known-sites
isolate_name.raw_variants.vcf –O isolate_name.recal_data.table
gatk ApplyBQSR –R reference.fa –I isolate_name.sorted.marked.bam –-bqsr-recal-file
isolate_name.recal_data.table –O isolate_name.recal_reads.bam
```

- Do this again (be careful with file names!) to acquire final BAM file with your final alignment

Is the alignment any good?

Quality control

- Two basic ways to check the quality of your alignment at this stage:
  - Coverage
  - Mapping statistics
- Coverage: how many reads map to a certain area
  - Useful to see as a good alignment has an even coverage across the whole genome in order to accurately/confidently call variants.
  - Depth of coverage can be used to observe copy number variation (CNV) or ploidy events

```
picard CollectWgsMetrics -R reference.fa -I isolate_name.post_recal_reads.bam -O
isolate_name.metrics.txt
```

```
samtools depth isolate_name.post_recal_reads.bam > isolate_name.coverage.txt
```

Is the alignment any good?

Coverage

- Samtools command gives you per base coverage (useful for CNV detection)
- Picard command gives a summary
  - Average coverage
  - Number of positions with zero coverage

Is the alignment any good?

Mapping stats

- Samtools
  - ‘flagstat’ command
  - Want:
    - a high percentage of mapped reads (>95%)
    - a high percentage of properly paired (if paired end reads)
    - low percentage of singletons

```
7417232 + 0 in total (QC-passed reads + QC-failed reads)
287618 + 0 duplicates
4534962 + 0 mapped (61.14%:-nan%)
7417232 + 0 paired in sequencing
3708616 + 0 read1
3708616 + 0 read2
4528278 + 0 properly paired (61.05%:-nan%)
4534962 + 0 with itself and mate mapped
```